

ASSIGNMENT 3

Course: B.Tech. IV year VII Sem Cryptography and Network Security (BCSE-1004)

Number of Questions: 10

Number of days for Submission: 7

1. A financial institution wants to secure its online banking transactions. They are considering using MACs. Explain how a Message Authentication Code works and provide a specific example of how it could be used in online banking to ensure the integrity and authenticity of transactions.

2.(i) A company wants to implement digital signatures for email communication to prevent phishing attacks. Explain how digital signatures can help in this context and how recipients can verify the authenticity of signed emails.

(ii) You work for a large organization that uses a KDC for authentication. An employee reports that they can't access their account. Describe the steps you would take to troubleshoot the issue and ensure the KDC is functioning correctly.

3. A user complains that they frequently have to re-enter their credentials throughout the day. Explain the concept of Kerberos tickets and ticket expiration. How can you adjust the Kerberos ticket lifetime to improve the user experience while maintaining security?

4. John is a cybersecurity analyst working for a large financial institution. He needs to communicate sensitive financial data and reports to his team members and clients regularly. The company's policy mandates that all email communications involving financial data must be encrypted to protect against unauthorized access and ensure data integrity. In this scenario, John decides to use PGP to secure his email communications.

a. How does John apply PGP to secure his email communications in this scenario?

b. What are the benefits of using PGP in this scenario?

c. Are there any potential challenges or considerations when using PGP for secure email communication in this scenario?

5. Describe Diffie-Hellman key exchange algorithms. User A and B use Diffie-Hellman key exchange technique with a common prime $q = 71$ and a primitive root $a = 7$.

1) If user A has private key $X_A = 5$, What is A's public key Y_A

2) If user B has private key $X_B = 12$, What is B's public key Y_B

3) What is the shared secret key

6. Explain the general format of an X.509 Certificate. How is an X.509 Certificate revoked?

7. An e-commerce website is experiencing frequent Distributed Denial of Service (DDoS) attacks that disrupt its online business. Explain the concept of DDoS attacks, and suggest strategies and technologies to mitigate and prevent them.

8. Compare the calculation of the key material in SSL and TLS. Answer the following

a. Which calculation provides more security?

b. Which calculation is more efficient in term of space and time?

9. Harry wants to implement end-to-end encryption in his messaging application. What are the steps involved in setting up end-to-end encryption, and what challenges might he encounter in ensuring user privacy while still providing a seamless messaging experience?

10. In an Intrusion detection scheme, false positives and negatives are an integral part. Justify whether you will go for a scheme with high false positive but less false negative or the other way round.

ASSIGNMENT 4

Course: B.Tech. IV year VII Sem Cryptography and Network Security (BCSE-1004)

Number of Questions: 10

Number of days for Submission: 7

1. A company's employees have fallen victim to phishing attacks in which attackers impersonate trusted entities to steal sensitive information. How can the company enhance its web security to prevent such attacks? Provide specific measures and technologies that can be implemented.
2. In a corporate environment, Jack needs to establish a secure VPN connection to access sensitive company resources remotely. Describe the cryptographic protocols and technologies that can be used to ensure the confidentiality and integrity of data transmitted over the VPN.
3. You are the IT administrator for a medium-sized company, and you have been tasked with improving the security of your organization's email communication. You need to implement S/MIME (Secure/Multipurpose Internet Mail Extensions) for email encryption and digital signatures, as well as enhance virus detection for incoming and outgoing emails. Please provide a step-by-step plan for implementing S/MIME and improving virus detection in your organization's email system.
4. What are the requirements of Kerberos? Differentiate between Kerberos version 4 and version 5. What are the shortcomings of Kerberos 4? How are they overcome in Kerberos version 5?
5. Lisa is concerned about the security of her mobile device and wants to enable encryption. How can she encrypt the data on her smartphone, and what are the potential benefits and drawbacks of doing so?
6. John is a cybersecurity analyst working for a large financial institution. He needs to communicate sensitive financial data and reports to his team members and clients regularly. The company's policy mandates that all email communications involving financial data must be encrypted to protect against unauthorized access and ensure data integrity. In this scenario, John decides to use PGP to secure his email communications.
 - a. How does John apply PGP to secure his email communications in this scenario?
 - b. What are the benefits of using PGP in this scenario?
 - c. Are there any potential challenges or considerations when using PGP for secure email communication in this scenario?

7. Show how SSL or TLS reacts to a replay attack. That is, show how SSL or TLS responds to an attacker that tries to replay one or more handshake messages.

8. Describe the component of a virus code. Explain the purpose of the components of the virus. How does the virus protect itself from being detected by anti-virus software?

9. A small e-commerce company, "ShopSafe," has been experiencing a series of security incidents that involve unauthorized access to their customer database. The company's IT team suspects that the breach might be related to a firewall misconfiguration. As a network security consultant, you have been called in to assess the situation.

You are tasked with conducting a comprehensive review of ShopSafe's firewall configuration to identify any weaknesses or misconfigurations that could have led to the recent security breaches. Please outline the steps you would take to assess the firewall and recommend improvements to enhance network security. Additionally, provide a brief explanation of why each step is essential in ensuring firewall effectiveness in cryptography and network security.

10. Briefly describe two possible approaches to proactive password checking. "Trojan horse is an imposter" Do you agree or disagree with this statement. Explain.